JEMS 2020 Virtual Conference

First-principles investigation of the impact of chemical order on the Magneto-crystalline anisotropy in $L1_0$-FeNi

Ankit Izardar, Claude Ederer

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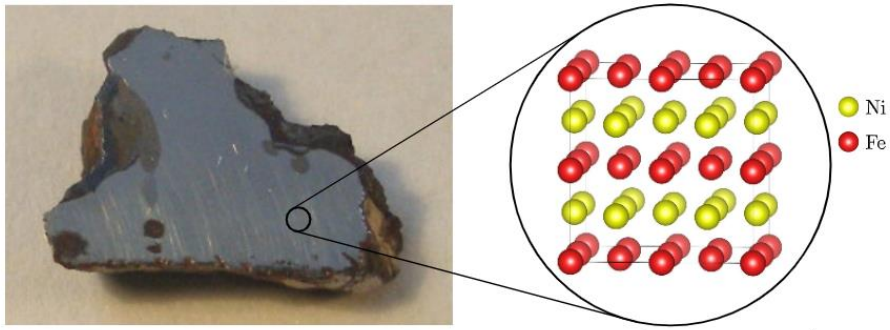
ankit.izardar@mat.ethz.ch

$L1_0$-$Fe_{50}Ni_{50}$



Iron meteorite (NWA 6259) $L1_0$ FeNi (Tetrataenite)

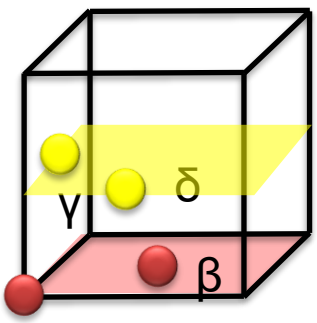
https://www.strahlen.org/tw/meteorite/meteorite_eisen5.html

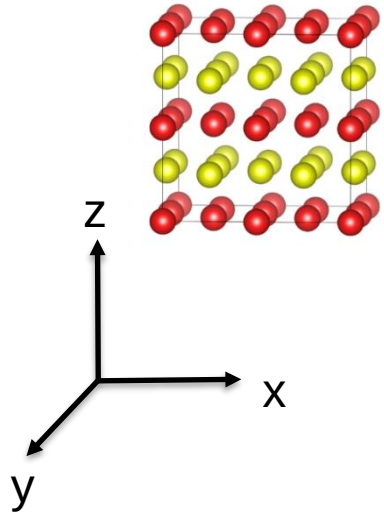
- Cheap and abundant constituent elements
- High Magneto-crystalline anisotropy and large saturation magnetization [1-2]
- Low order-disorder transition temperature (593 K)[3]
  - Laboratory synthesis challenging → difficult to fully characterize magnetic properties
- Previous studies[4-5] → necessary to consider both chemical and magnetic degrees of freedom

1. Pauleve et al., *J. App. Phys*. **39**, 989 (1968).
2. Lewis et al., *IEEE Magn. Lett*. **5**, 5500104 (2014).
3. Neel et al., *J. App. Phys*., **35**, 873 (1964).
4. Dang et al., *Phys. Rev. B*, **53**, 2291 (1996).
5. Lavrentiev et al., *Phys. Chem. Chem. Phys*, **16**, 16049 (2014).

Long range order parameter

Fcc lattice





Long range order parameter

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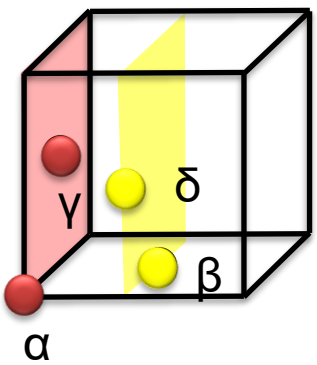
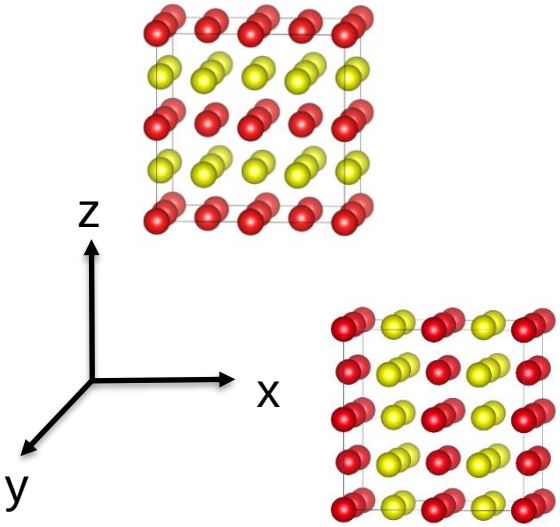
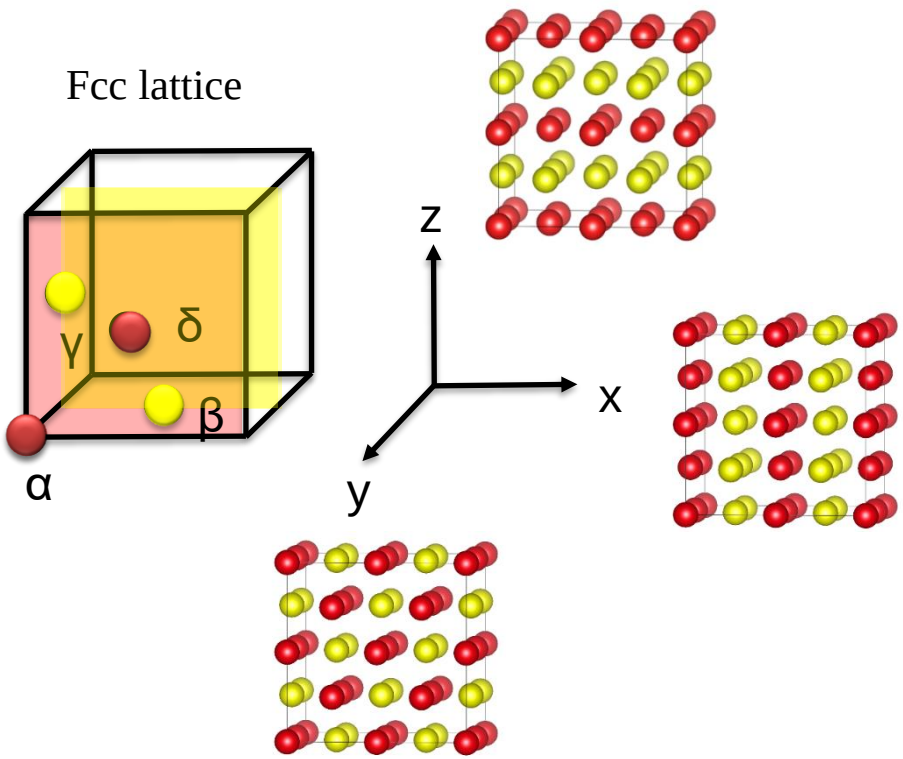




Long range order parameter



- Occupation probability

$$p_i^{Fe} = \frac{N_i^{Fe}}{N_i}, \quad i \in \alpha, \beta, \gamma, \delta$$

- Long range order

$$\boldsymbol{P_z = p_\alpha^{Fe} + p_\beta^{Fe} - 1}$$

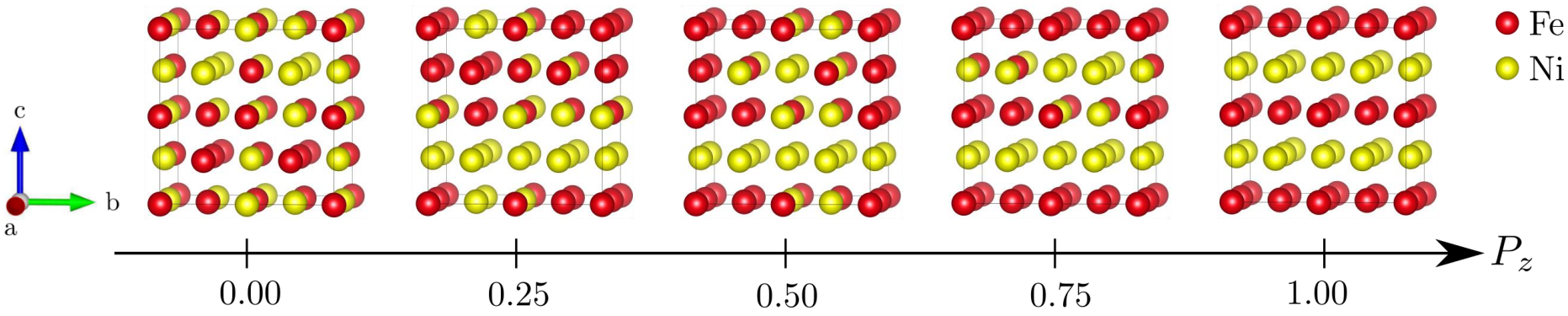
$$\boldsymbol{P_x = p_\alpha^{Fe} + p_\gamma^{Fe} - 1}$$

$$\boldsymbol{P_y = p_\alpha^{Fe} + p_\delta^{Fe} - 1}$$

Completely ordered $\rightarrow p_\alpha^{Fe} = p_\beta^{Fe} = 1,\ p_\gamma^{Fe} = p_\delta^{Fe} = 0 \rightarrow P_z = 1.0,\ P_x = 0,\ P_y = 0$

Completely disordered $\rightarrow p_i^{Fe} = 0.5 \rightarrow P_z = 0.0,\ P_x = 0,\ P_y = 0$

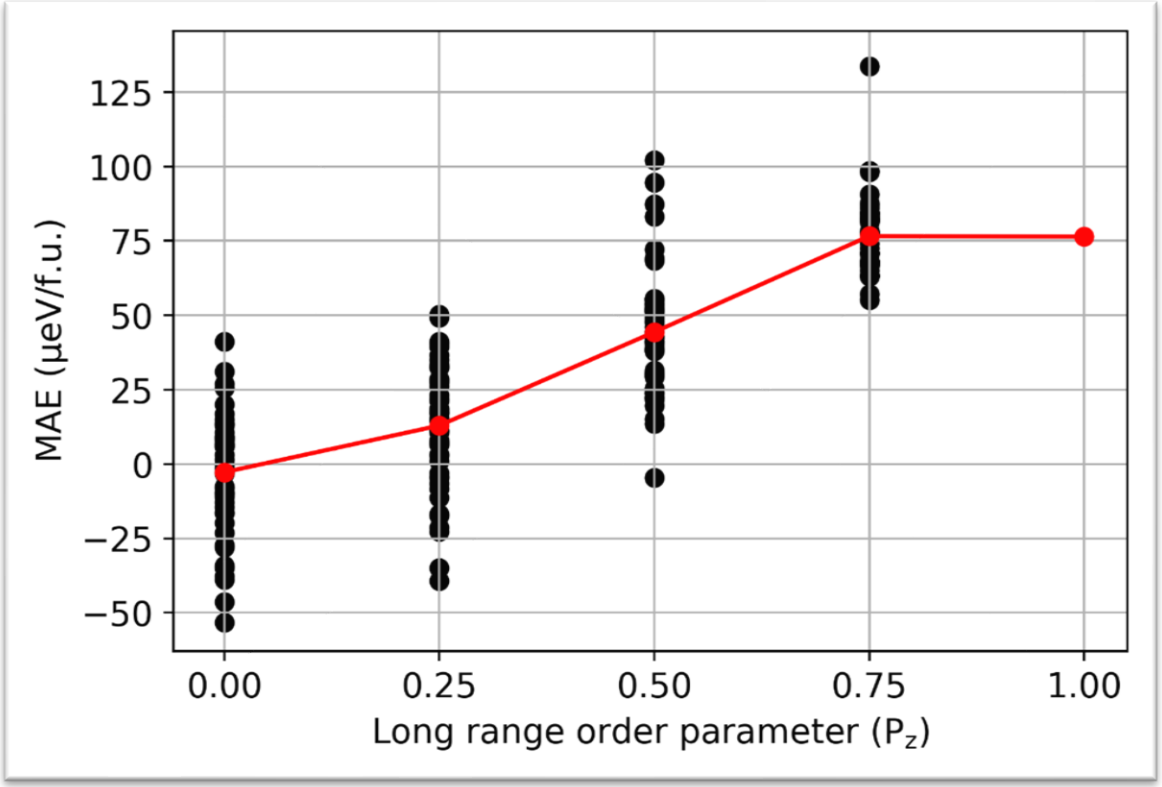
Supercell construction



- $2 \times 2 \times 2$ supercell of the conventional cubic cell
- For each $P_z < 1$ ($P_x = P_y = 0$), 50 random configurations are generated
- Total energy for each supercell calculated using DFT

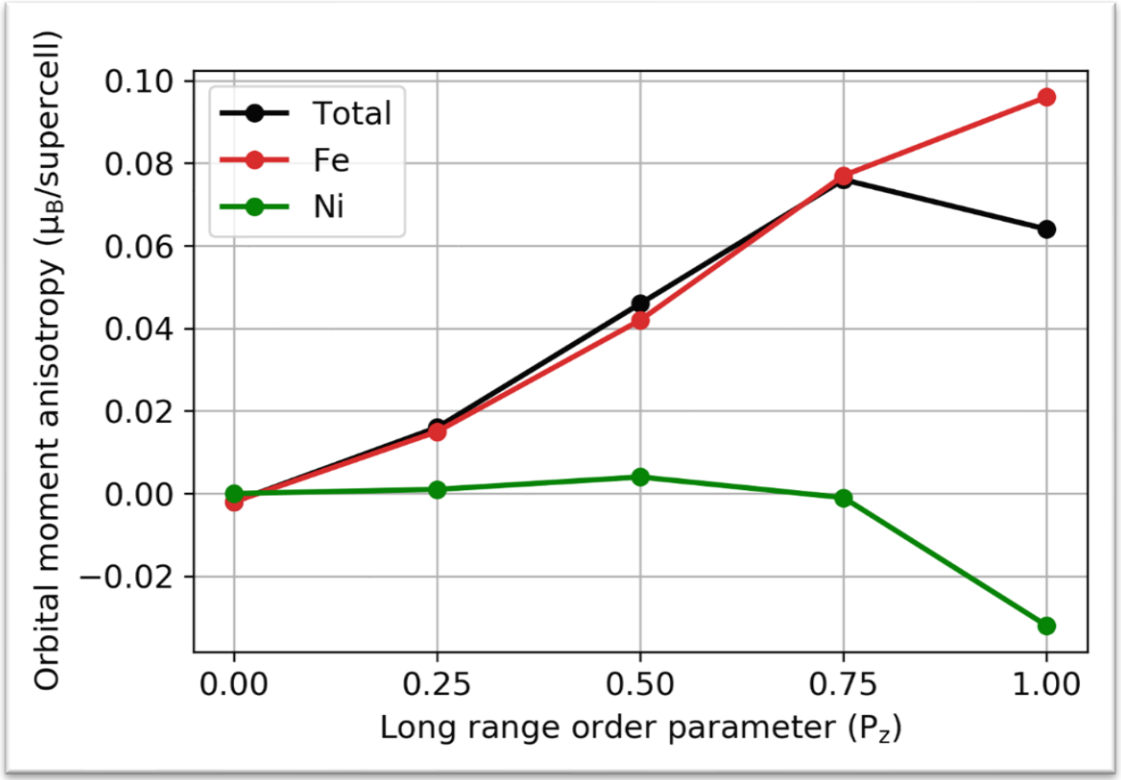
Magneto-crystalline anisotropy vs. order parameter

- $MAE = E_{M \parallel [100]} - E_{M \parallel [001]}$
- Positive MAE → Magnetic easy axis along [001]
- Sampling over 50 config. for each $P_z < 1$
- Maximum MAE already reached for $P_z = 0.75$



MAE does not decrease significantly if the deviations from the perfect order are not too large !

Orbital moment anisotropy (ΔL)

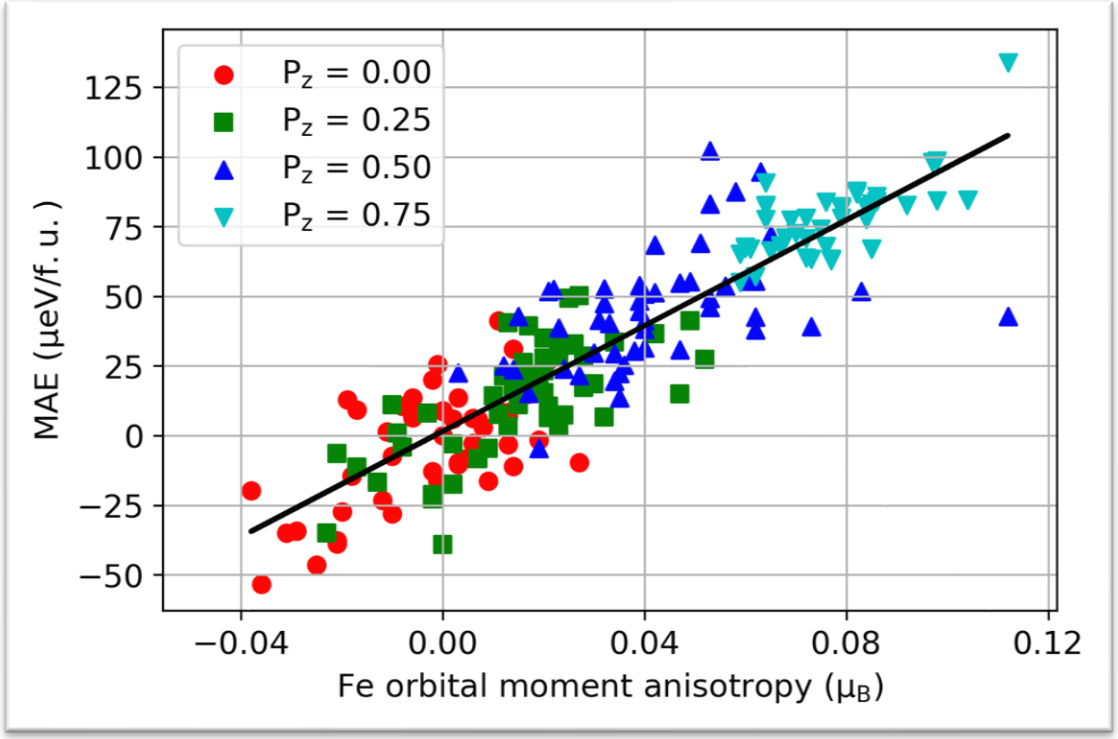


- Bruno's model[1]
  - MAE $\propto \Delta L$
- $\Delta L = L_{M \parallel [001]} - L_{M \parallel [100]}$
- Main contribution to the total ΔL for $P_z < 1$ comes from the anisotropy of Fe atoms

- For $P_z = 1$, orbital magnetic moment of Ni atoms are larger along [100] → Small decrease of ΔL from $P_z = 0.75$ to $P_z = 1$

1. P. Bruno, *Phys. Rev. B* **39**, 865 (1989)

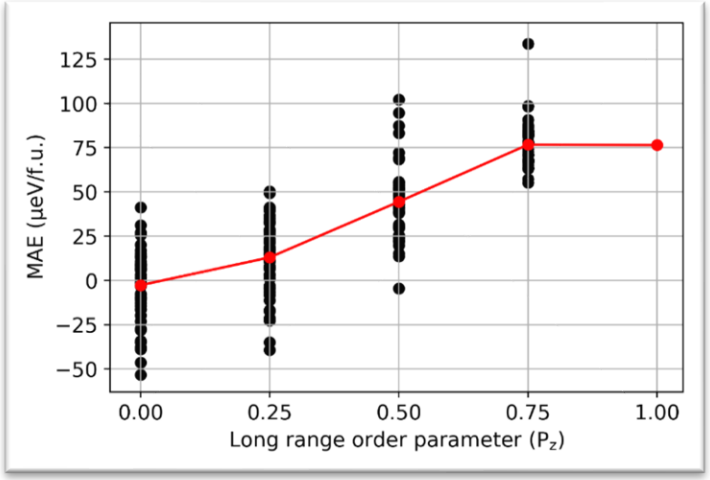
Correlation between MAE and ΔL



- Linear correlation between the quantities
- A noticeable spread on the level of the individual configurations
- Orbital anisotropy as a proxy for MAE → allows to analyze how the local environment affects both quantities

Summary & Conclusion

- Variation of MAE with degree of chemical order
  - Decrease not significant if deviations from perfect order are not too large
  - For $P_z = 0.75$, we find some config. with even higher MAE than fully ordered config.



- Correlation between MAE and orbital moment anisotropy
- Outlook
  - To investigate how the local chemical environment affect orbital moment anisotropy of atoms
  - Variation of Curie temperature with chemical order

PHYSICAL REVIEW MATERIALS **4**, 054418 (2020)

Interplay between chemical order and magnetic properties in $L1_0$ FeNi (tetrataenite): A first-principles study

Ankit Izardar and Claude Ederer
Materials Theory, ETH Zürich, Wolfgang-Pauli-Strasse 27, 8093 Zürich, Switzerland

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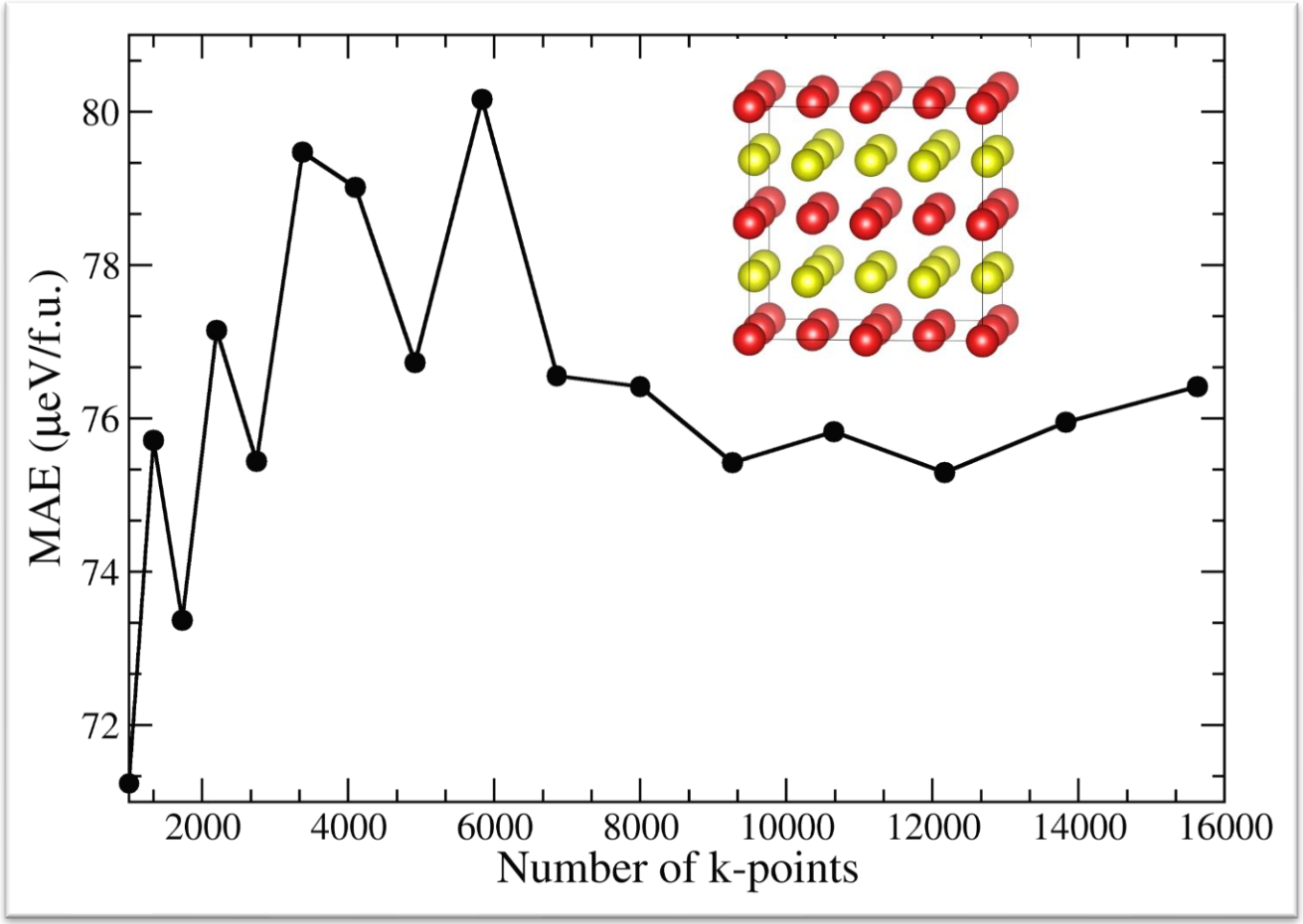
We use first-principles-based calculations to investigate the interplay between chemical order and the magnetic properties of $L1_0$ FeNi. In particular, we investigate how deviations from perfect chemical order affect the energy difference between the paramagnetic and ferromagnetic states as well as the important magnetocrystalline anisotropy energy. Our calculations demonstrate a strong effect of the magnetic order on the chemical order-disorder transition temperature and, conversely, a strong enhancement of the magnetic transition temperature by the chemical order. Most interestingly, our results indicate that the magnetic anisotropy does not decrease significantly as long as the deviations from perfect order are not too large. Moreover, we find that in certain cases a slight disorder can result in a higher anisotropy than for the fully ordered structure. We further analyze the correlation between the magnetocrystalline anisotropy and the orbital magnetic moment anisotropy, which allows to study the effect of the local chemical environment on both quantities, potentially enabling further optimization of the magnetocrystalline anisotropy with respect to chemical order and stoichiometric composition.

DOI: 10.1103/PhysRevMaterials.4.054418

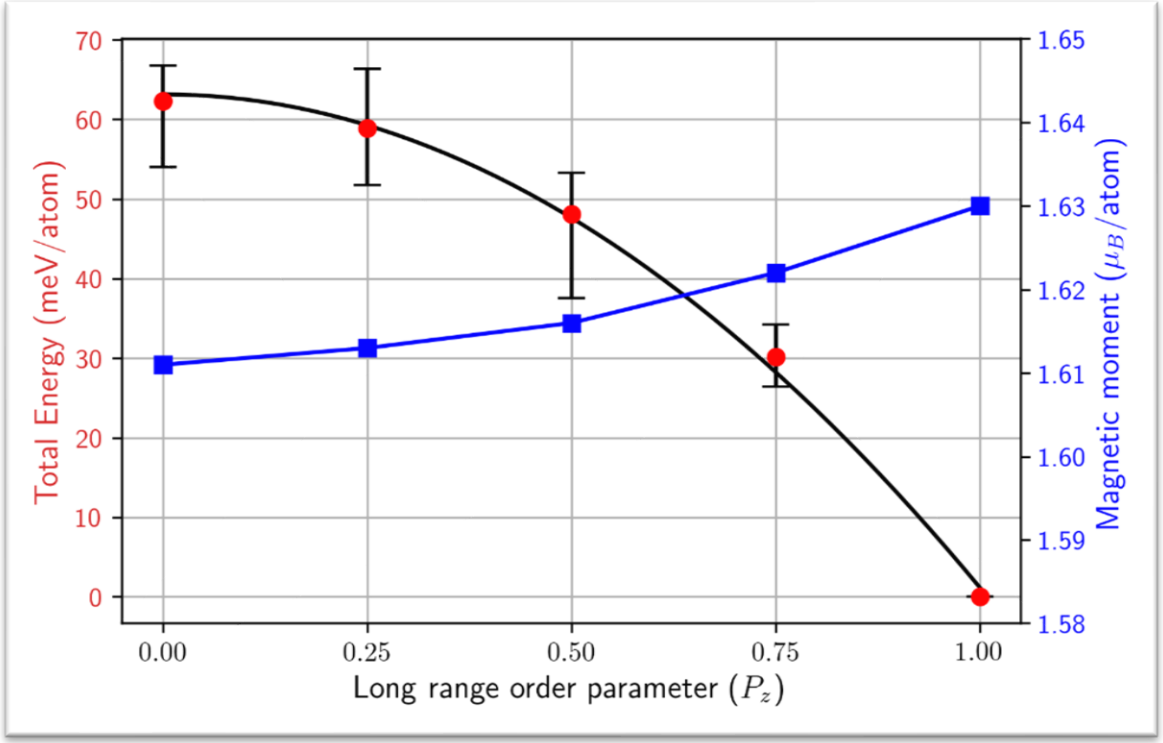
Izardar et al., *Phys. Rev. Mat.* 4, 054418 (2020)

Thank you !

MAE convergence for ordered configuration



Total energy vs. Order parameter

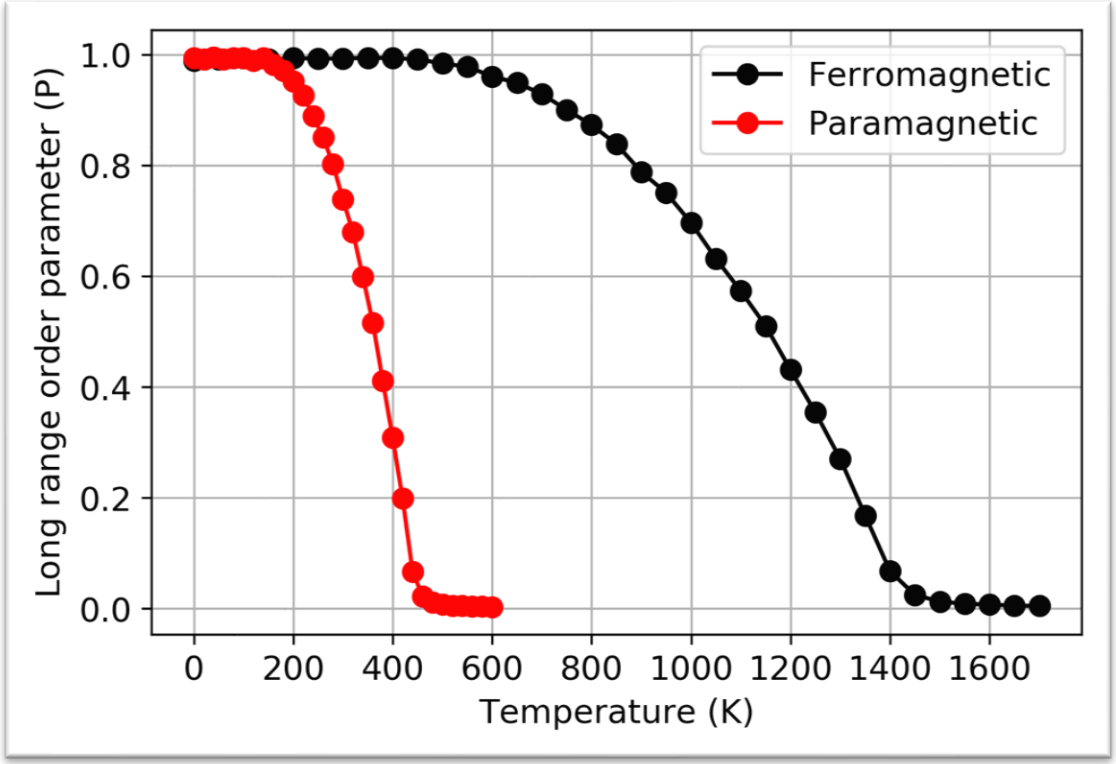


- Averaged total energies are well fitted by $E = E_0 - \Delta E P_z^2$
- Good quality of quadratic fit → accurate mean energies can be obtained by using 50 configurations for each $P_z < 1$

- Average total magnetic moment depends weakly on P_z

Temperature dependence of LRO

- $T_{o-d}^{Ferromag.} \sim 1400\ K$
- Higher than $T_{o-d}^{exp.} \sim 593\ K$
- What about paramagnetic state ?
- Modeling of paramag. state → Disordered local moment method (DLM)
- $T_{o-d}^{paramag.} \sim 450\ K$



Strong coupling between chemical o-d transition and the magnetic state in $L1_0$-FeNi !